# Curriculum Vitae

## Dr. Dennis Ulbrich

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Mathematician with research focus in nonlinear analysis, dynamical systems and kinetic theory; several years of teaching experience in undergraduate mathematics (including large-scale service courses such as "Mathematik 1"). Experienced in teaching, student supervision, and academic course organization.

## Teaching experience

- **Mathematik 1** — Bremen
  *Hochschule Bremen, School of Nature and Engineering* — *10.2024 – 3.2025*
  - Lecture and tutorial, approx. 50 participants
  - Creation and correction of written exams
  - Creation of weekly exercise sheets
  - Organisation via online platform "AULIS"
- **Finite Mathematics** — Bremen
  *Jacobs University Bremen, Mathematical Sciences* — *1.2022 – 6.2022*
  - Lecture and tutorial, approx. 50 participants
  - Creation and correction of written exams
  - Creation of weekly exercise sheets
  - Organisation via Moodle
- **Introduction to Dynamical Systems** — Bremen
  *Jacobs University Bremen, Mathematical Sciences* — *1.2022 – 6.2022*
  - Proseminar, approx. 15 participants
  - Providing literature and further material
  - Grading final presentations of the participants

## Research interests

ODE and PDE and their dynamics, nonlinear waves, dynamical systems, ergodic theory, kinetic theory and entropy methods, Boltzmann equation and its BGK type approximations, long time behavior of solutions (hypocoercivity), physical applications, models of chemical reactions

## Employment

- **Research assistant** Bremen
  *Hochschule Bremen, School of Electrical Engineering and Computer Science* *2.2025 – 5.2025*
  - Project work: AI-based transmission, analysis, verification of handwritten documents
- **Lecturer in Mathematics** Bremen
  *Hochschule Bremen, School of Nature and Engineering* *10.2024 – 3.2025*
  - Mathematik 1 (cf. Teaching experience)
- **Postdoctoral researcher** Münster
  *Universität Münster, Fachbereich Mathematik, Institut f. Analysis u. Numerik* *6.2023 – 6.2024*
  - Research on discrete hypocoercivity within DFG project 456849348
  - Supervision: Prof. Dr. Marlies Pirner
  - Keywords:
    * Hypocoercivity
    * Kinetic equations
    * BGK-type approximations
    * Chemical reactions
    * Entropy methods
- **IT-Consultant** Affinis AG Bremen
  *Professional internship* *9.2022 – 2.2023*
- **Lecturer in Mathematics** Bremen
  *Jacobs University Bremen* *1.2022 – 6.2022*
  - Finite Mathematics, Introduction to Dynamical Systems (cf. Teaching experience)
- **Research assistant (PhD student)** Bremen
  *University of Bremen, Department of Mathematics* *4.2017 – 9.2021*
  - Research within DFG project RA 27288/1-1
  - Research groups: Nonlinear Analysis and Applied Analysis, Stochastics and Dynamical Systems
  - PhD thesis: *Ergodic theory of nonlinear waves in discrete and continuous excitable media*
    * *Advisors:* Prof. Dr. Jens Rademacher (University of Hamburg), Prof. Dr. Marc Keßeböhmer (University of Bremen)
    * *Referees:* Prof. Dr. Jens Rademacher, Prof. Dr. Ian Melbourne (Warwick University)
  - Keywords:
    * Nonlinear analysis
    * Ergodic Theory
    * PDE
    * Excitable media
    * Cellular automata
    * Dynamical Systems

## Education

- **Dr. rer. nat., *Mathematics*** Bremen
  *University of Bremen, Department of Mathematics* *4.2017 – 9.2021*
  - PhD thesis:
    *Ergodic theory of nonlinear waves in discrete and continuous excitable media*
  - Advisors: Prof. Dr. Jens Rademacher (University of Hamburg), Prof. Dr. Marc Keßeböhmer (University of Bremen)
  - Referees: Prof. Dr. Jens Rademacher, Prof. Dr. Ian Melbourne (Warwick University)
  - cf. MGP
- **Brückenstipendium** Bremen
  *University of Bremen, Department of Mathematics* *4.2016 – 4.2017*
  - Further studies
  - Extensions of the results of my Master's thesis
  - Successful acquisition of third-party funding (DFG) for my PhD project
- **M.Sc. and B.Sc., *Mathematics*** Bremen
  *University of Bremen* *10.2011 – 4.2016*
  - M.Sc. thesis:
    *Dynamics of the three-state 1D Greenberg-Hastings cellular automaton*
    * Referees: Prof. Dr. Jens Rademacher, Dr. Tony Samuel (University of Exeter)
  - B.Sc. thesis:
    *Unerwartete Fehler bei bedingten Erwartungswerten und Wahrscheinlichkeiten*
    * Referees: Prof. Dr. Werner Brannath (University of Bremen), Dr. Kurt Falk (University of Kiel)

## Publications

**Peer-reviewed journal articles:**

- A. Pauthier, J.D.M. Rademacher, **D. Ulbrich**.
  Weak and strong interaction of excitation kinks in scalar parabolic equations.
  J Dyn Diff Equat. Published: 30 July 2021; Volume 35, pages 2199-2235, (2023), [DOI] [arXiv]
- M. Keßeböhmer, J.D.M. Rademacher, **D. Ulbrich**.
  Dynamics and topological entropy of 1D Greenberg-Hastings cellular automata.
  Ergodic Theory and Dynamical Systems. 2021;41(5):1397-1430, [DOI] [arXiv]

**Ongoing work:**

- L. Liu, M. Pirner, **D. Ulbrich**.
  Discrete hypocoercivity for a nonlinear kinetic reaction model without initial close-to-equilibrium assumption.

**Thesis:**

- D. Ulbrich.
  Ergodic theory of nonlinear waves in discrete and continuous excitable media.
  Diss., 2021, [DOI]

## Academic services/ academic responsibilities

**Supervision:**

- Student research project, University of Bremen, Department of Mathematics, 2021 – 2022

**Organisation:**

- Supporting several summer and winter schools, University of Bremen, 2016 – 2021
- Maintaining the Mathematical Collection of the University of Bremen and creating its website

## Grants

- Brückenstipendium, University of Bremen, Department of Mathematics, 4.2016 – 4.2017

## IT skills

- LaTeX: advanced
- Matlab, Mathematica, Java, C++, Oracle PL/SQL: basic

## Languages

- German: native speaker
- English: fluent

updated: Mar 20, 2026